Free Energy: An Honest Audit of Autoregressive Language Models and the Energy-Based Alternative

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Abstract

Autoregressive language models dominate sequence modeling, yet a long-running set of theoretical objections holds that they are fundamentally limited: local next-token prediction cannot scale to global correctness; softmax heads cap representational rank; training under maximum likelihood is mode-covering; and per-token errors compound to make long, exact sequences asymptotically unattainable. Most critiques bundle three distinct design choices — the autoregressive factorization, the softmax parameterization, and the maximum-likelihood objective — into one indictment. We separate them. We catalog the standard objections, tag each by the layer it actually constrains (architecture, objective, trained behavior, or external resource), and rate which are theorems, which are robust empirical findings, and which depend on premises modern systems explicitly violate. The most-cited compounding bound $(1-\varepsilon)^T$ is mathematically valid but predicates on independent, constant-rate, unrecoverable errors; verifiers, tools, and search are precisely the system features designed to break those premises. By contrast, the softmax bottleneck and mode-covering pressure are genuine theorems, and the partition-function obstacle of energy-based models is a real cost rather than a free escape. We then survey the alternatives — energy-based models, score-matching and noise-contrastive estimation, denoising and masked diffusion, joint-embedding predictive architectures, and verifier-augmented decoding — and ask, for each, *where the hard part lives*: local prediction with external verification, or global compatibility scoring with hard inference and normalization. This work is the literature-grounded half of a two-part project; a companion empirical suite (`experiments/02_ar_pros_cons`) tests each surviving claim end-to-end on a formal-verification domain with a hard checker.

1 Introduction

The discourse around autoregressive language models has settled into two slogans. One says they work, scale, and only need more compute, data, and tools. The other says they are fundamentally broken because each generated token has a nonzero probability of being wrong, and a length-T generation is fully correct with probability at most $(1 - \varepsilon)^T$, which decays exponentially.

Both slogans are too clean.

The first ignores that local fluency is not the same as global correctness on tasks like proof writing, program synthesis, or long-horizon planning. The second ignores that the math is valid only under three premises: independent token errors, a constant per-token error rate, and unrecoverability of any single wrong token. Modern deployed systems are designed precisely to violate those premises. Tools catch syntax, arithmetic, and type errors; verifiers reject invalid proofs; search lets a model

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retry; and reinforcement learning from verifier feedback shifts the empirical error process away from the i.i.d. assumption the bound requires.

Most critiques of autoregressive language models bundle three distinct design choices into a single object. The first is the *autoregressive factorization* $p(x) = \prod_t p(x_t \mid x_{<t})$, which trades the global partition function of a sequence-level model for T local normalizers. The second is the *softmax parameterization* of each conditional, which factors as an exponential over a learned hidden state followed by normalization over the vocabulary. The third is the *maximum-likelihood objective with teacher forcing*, which trains each conditional on clean prefixes drawn from data rather than from the model's own past samples. These three choices are independent. Different objections indict different choices, and any honest case for an alternative architecture must specify which of the three it is replacing.

The contribution of this paper is a layer-disciplined map of the standard objections to autoregressive language models. We tag each objection by the layer at which it lives. *Architecture*-layer claims (e.g. the softmax bottleneck, pure-attention rank collapse, fixed compute per token) can be tested on a random-initialized model. *Objective*-layer claims (e.g. mode-covering pressure from forward-KL minimization) require a controlled training run. *Trained-behavior*-layer claims (e.g. error compounding, reversal failures, hallucination rates) require both a trained model and an evaluator with a notion of correctness. *External*-layer claims (e.g. the data wall) are properties of a resource curve, not of any single model. Mixing layers is the most common way to fool oneself about which mechanism is doing the work.

We then ask, for each objection, whether it is a theorem, a robust empirical finding, a weak empirical finding, or refuted under modern conditions. Two clusters emerge. The softmax bottleneck (Yang et al., 2018), doubly-exponential rank collapse in pure attention (Dong et al., 2021), and forward-KL mode-covering pressure (Theis et al., 2016; Minka, 2005) are theorems whose preconditions are visible from the parameterization or the objective. The error-compounding bound (LeCun, 2022; Bachmann and Nagarajan, 2024), the reversal curse (Berglund et al., 2024), and most hallucination claims are empirical patterns whose realized effect under verifier-augmented systems is an open quantitative question rather than a fundamental impossibility.

Alternatives to the autoregressive recipe inherit, evade, or relocate the same costs. Energy-based models (LeCun et al., 2006; Grathwohl et al., 2020) score whole objects globally and avoid the local-then-global mismatch, but pay for the partition function or its approximations at training and at inference. Score matching (Hyvärinen, 2005), noise-contrastive estimation (Gutmann and Hyvärinen, 2010), and ratio matching (Hyvärinen, 2007) sidestep Z_θ during training but typically push the cost to inference-time sampling. Diffusion and denoising models (Sohl-Dickstein et al., 2015; Ho et al., 2020; Song et al., 2021b; Nichol and Dhariwal, 2021) score a hierarchy of perturbations to the data and yield a tractable training loss at the cost of an iterative inference loop. Discrete and masked diffusion (Lou et al., 2024; Nie et al., 2024) bring the same idea to vocabulary-indexed sequences. Joint-embedding predictive architectures (LeCun, 2022) replace per-token reconstruction with feature-space prediction. Each is a different choice about *where the hard part lives*.

This paper is the literature-grounded half of a two-part project. The companion experiment, `experiments/02_ar_pros_cons` in the same repository, tests each surviving claim with an isolated probe at the correct layer and then re-measures the same mechanism inside an end-to-end run on a domain with a hard verifier (Lean 4, via VeriBench). We refer to that suite throughout, both for which claims it pre-registers as load-bearing and for which it pre-registers as likely masked once everything is trained together.

The remainder of the paper is organized as follows. Section 2 restates the honest case for autoregressive language models: the four design properties that explain why they scaled. Section 3 catalogs the standard objections, with layer tags. Section 4 evaluates each objection against the bar of theorem, robust empirical, weak empirical, or refuted, and identifies the three independent indictments the surviving claims support. Section 5 introduces energy-based models from the global-scoring viewpoint and isolates the partition-function obstacle as the central cost. Sections 6 and 7 survey the training and inference methods that aim to evade or amortize that cost. Section 8 surveys non-EBM alternatives. Section 9 discusses where the hard part should live and the empirical questions that this paper deliberately defers to the companion suite.

2 What Autoregressive Language Models Get Right

Any honest critique of autoregressive language models has to begin with what they bought. The factorization $p(x_1, \ldots, x_T) = \prod_t p(x_t \mid x_{<t})$ enabled four properties that, taken together, explain a decade of scaling progress.

Tractable normalized likelihood. For an energy-based model over a length-T sequence with vocabulary $\mathcal{V}$, the natural probability is

$$p_\theta(x) \;=\; \frac{\exp(-E_\theta(x))}{Z_\theta}, \qquad Z_\theta \;=\; \sum_{x \in \mathcal{V}^T} \exp(-E_\theta(x)). \tag{1}$$

The normalizer in Equation (1) sums over $|\mathcal{V}|^T$ sequences for discrete vocabularies and becomes a high-dimensional integral over $(\mathbb{R}^d)^T$ in the continuous-embedding case. Without special structure on E_θ, exact normalization is intractable. The autoregressive factorization replaces this single global normalizer with T local normalizers, each summing over $|\mathcal{V}|$ tokens:

$$p_\theta(x) \;=\; \prod_{t=1}^{T} p_\theta(x_t \mid x_{<t}), \qquad \sum_{x_t \in \mathcal{V}} p_\theta(x_t \mid x_{<t}) \;=\; 1. \tag{2}$$

Summing out the joint from the right confirms normalization by construction. This is not a minor implementation detail. It is the architectural reason an exact likelihood is available for end-to-end gradient training, and it is the reason maximum-likelihood was a viable training objective at the scale of modern corpora.

Dense supervision per token. A length-T sequence furnishes T next-token prediction losses. No human label is required. Combined with the chain rule in Equation (2), this makes every position in a corpus a training example, and it is one reason gradient-based pretraining on web-scale text yielded usable models at all (Radford et al., 2018; Brown et al., 2020; Kaplan et al., 2020).

Teacher forcing decouples training from inference dynamics. At training time, each conditional is evaluated on the true prefix from the data rather than on the model's own past samples. The target is local: a categorical distribution over $|\mathcal{V}|$ tokens given $x_{<t}$. This makes the optimization landscape close to that of independent classification problems, one per position, and avoids the variance and instability of bootstrapping from the model's own samples. The cost — a discrepancy between training and free-running deployment, sometimes called *exposure bias* (Bengio et al., 2015; Ranzato et al., 2016) — is real but smaller than initially feared on large models, and we revisit it as a candidate trained-behavior claim in Section 3.

Streaming generation as a universal interface. Left-to-right factorization lets generation begin before the full output is determined. The token stream serves as a common substrate for prose, code, proof steps, structured outputs, tool calls, and intermediate reasoning traces. This is the engineering interface that allowed chain-of-thought (Wei et al., 2022), tool use (Schick et al., 2023), and verifier-in-the-loop pipelines (Cobbe et al., 2021; Lightman et al., 2024) to compose without new training paradigms.

The factorization itself is not an expressivity bottleneck. By the chain rule of probability, any joint distribution over a finite sequence space admits a left-to-right factorization with sufficiently expressive conditionals. The practical bottlenecks are model capacity, training data, the objective, and inference; the existence of the factorization is not a proof of weakness. Many empirical failures attributed to "autoregression" more accurately indict the parameterization (softmax over a low-rank hidden state), the objective (mode-covering MLE), or the inference procedure (one-pass blind rollout) rather than the chain-rule decomposition itself. We return to this disentanglement repeatedly.

These four properties — tractable likelihood, dense supervision, teacher forcing, and streaming — make up the case the rest of the paper is graded against. A successful alternative architecture must either (i) replicate them, (ii) supply something strictly more useful in their place, or (iii) make a clearly bounded trade in a specific regime (long-horizon exactness, global coherence, adaptive compute) where the gain is large enough to justify the loss.

3 The Standard Objections, Cataloged by Layer

We catalog the objections that recur in the literature and on social media, organized by the layer at which they live. This is the input to Section 4, which judges each.

3.1 Architecture-layer claims

Architecture-layer claims can be tested on a random-initialized model with no training data. A positive control should show the effect at the limit predicted by the theory; if it does not, the probe is broken, not the claim.

(A1) **Softmax bottleneck.** A softmax head over a d-dimensional hidden state can produce a log-probability matrix of rank at most $d+1$. If the true next-token log-probability matrix has rank greater than $d+1$, the model class cannot represent it (Yang et al., 2018).

(A2) **Doubly-exponential rank collapse in pure attention.** A stack of pure self-attention layers (no residual connections, no MLPs, no layer normalization) converges to a rank-one representation doubly-exponentially in depth (Dong et al., 2021). Residuals and MLPs slow but do not eliminate the underlying contraction.

(A3) **Locally normalized softmax is a per-step partition function.** Each softmax layer evaluates $Z = \sum_k \exp(q^\top k_k/\sqrt{d})$ over its candidate set; the output head evaluates $\sum_v \exp(w_v^\top h)$ over the vocabulary. Ramsauer et al. (2021) show that softmax attention is one gradient step on a continuous Hopfield energy whose specific exponential form was chosen to make Z tractable. The cost is paid at every layer and every position.

(A4) **Fixed compute per token is a representational ceiling.** Standard transformers perform $O(1)$ sequential operations per generated token. Circuit-complexity results (Merrill and Sabharwal, 2023; Strobl et al., 2024) place transformer expressivity in low-depth uniform classes, implying problems that require more sequential depth cannot be solved in a single forward pass. Chain-of-thought externalizes serial computation into the token stream and is, on this reading, evidence of the limit rather than a refutation of it.

(A5) **Fixed hidden-state capacity bounds in-context memory.** Architectures with a bounded hidden state cannot copy strings of arbitrary length (Jelassi et al., 2024; Arora et al., 2024). Self-attention's key/value cache sidesteps this by growing with context; state-space models with fixed-dimensional state inherit a sharper version of the same constraint.

3.2 Objective-layer claims

Objective-layer claims are about the geometry of the training loss. They require a controlled training run to test the realized effect, but the underlying mechanism is derivable from the loss alone.

(O1) **Mode-covering pressure of maximum likelihood.** Maximum-likelihood estimation minimizes the forward KL $D_{\mathrm{KL}}(p_{\mathrm{data}} \,\|\, p_\theta)$, which diverges if p_θ places near-zero mass on any datum the data distribution supports (Theis et al., 2016; Minka, 2005). The objective therefore presses the model to cover every observed mode, including noise, contradictions, and rare or undesired patterns.

(O2) **Teacher forcing differs from free-running generation.** At training time, conditionals are evaluated on data prefixes; at inference time, the model conditions on its own past samples. Ranzato et al. (2016); Bengio et al. (2015) formalize the resulting distribution shift, often called exposure bias.

(O3) **Next-token likelihood and global correctness are different metrics.** A proof, a program, or a multi-step plan can have low cumulative log-likelihood and high correctness, or high log-likelihood and zero correctness. Optimizing one is not optimization of the other.

3.3 Trained-behavior-layer claims

Trained-behavior claims involve the dynamics of a trained model under inference, including its interaction with a verifier or environment. These claims are empirical: they require both a trained model and an evaluator with a notion of correctness.

(T1) **Error compounding: the $(1-\varepsilon)^T$ argument.** Under the assumptions that (i) each generated token is wrong with constant probability ε, (ii) token errors are independent across positions, and (iii) any wrong token is unrecoverable, the probability that a length-T output is fully correct is $(1-\varepsilon)^T$, which decays exponentially in T (LeCun, 2022; Bachmann and Nagarajan, 2024).

(T2) **Reversal curse.** A model trained on the statement "A is B" does not reliably infer "B is A" (Berglund et al., 2024). The pattern survives across scales and is presented as evidence that next-token prediction encodes directional co-occurrence rather than symmetric relations.

(T3) **Hallucination.** Trained autoregressive models confidently produce factually unsupported continuations. Kalai and Vempala (2024) argue calibrated next-token training induces a structural pressure toward plausible completion; Xu et al. (2024) and Kandpal et al. (2023) catalog empirical patterns. The strong form of the claim is that hallucination is *inevitable*.

(T4) **Brittleness and the Lipschitz–margin gap.** For a model with global Lipschitz constant L, the input-space margin obeys $\|\delta\| \geq f_\theta(x)/L$, where $f_\theta(x)$ is the unnormalized output gap and $L \leq \prod_i \|W_i\|_2$ (Tsuzuku et al., 2018; Virmaux and Scaman, 2018). Output confidence can therefore be inflated by scaling weights while the geometric margin stays small, producing brittleness under small input perturbations.

3.4 External-layer claims

External-layer claims are not about any single model. They are about the resource curve on which the field is operating.

(E1) **Data wall.** High-quality public text is finite and is being consumed (Villalobos et al., 2024). Combined with empirical neural scaling laws (Kaplan et al., 2020; Hoffmann et al., 2022) that require exponential data for linear loss improvements, the cheap-gains regime is expected to end before the loss has saturated to its irreducible floor. The claim is not that scaling stops working; it is that the implicit dataset assumption behind the scaling laws breaks before the curve does.

3.5 Three independent indictments

The catalog above bundles three logically independent design choices. The factorization, the parameterization, and the objective each take some of the heat. We will repeatedly need the following partition:

Against the softmax / local normalization: (A1), (A2), (A3). These motivate energy-based heads, mixture-of-softmaxes, or non-normalized attention substrates.

Against the MLE objective: (O1), (T3) (in its calibrated form). These motivate noise-contrastive estimation, margin or contrastive losses, and reinforcement learning from verifier rewards.

Against autoregressive generation under fixed compute: (A4), (T1), (T4). These motivate inference-as-optimization (energy descent, iterative refinement), search-augmented decoding, and verifier-in-the-loop pipelines.

(A5), (O2), (O3), (T2), and (E1) cross the three buckets or sit outside them.

The point of separating these is that an alternative architecture only addresses the indictments it actually targets. An energy-based head with a softmax-trained encoder still pays the cost of mode-covering MLE. A non-softmax attention substrate still pays the cost of fixed compute per token. The most-cited objection in popular discussion, (T1), indicts only the third bucket and only under premises that modern verifier-augmented systems explicitly violate; this is the subject of Section 4.

4 Which Objections Actually Hold

We now grade each claim from Section 3 against a clear bar: a claim is a *theorem* only if it follows from the math of the factorization, parameterization, or objective; a *robust empirical* finding only if it is reproduced across labs and scales with a stated mechanism; *weak empirical* if its support depends on a single setup or a fragile premise; and *refuted* if a direct counterexample exists under conditions a deployed system can satisfy. Table 1 previews the verdicts; the subsections justify each.

4.1 Theorems: rank, collapse, partition, compute, mode-covering

The strongest objections to the standard parameterization and objective are not conjectures. They follow from the math.

Table 1: Standing of each objection. "Layer" is from Section 3. "Verdict" is the standing of the claim under the bar of theorem / robust empirical / weak empirical / refuted / untested. "Bites" is what the surviving form of the claim indicts.

Tag	Claim	Layer	Verdict	Bites
(A1)	Softmax rank cap	architecture	theorem	softmax he
(A2)	Pure-attention rank collapse	architecture	theorem (pure); masked (trained)	softmax at
(A3)	Local partition function tax	architecture	established	softmax /
(A4)	Fixed compute ceiling	architecture	theorem	one-pass ge
(A5)	Fixed-state recall limit	architecture	theorem (SSMs); workaround (attn)	fixed-state
(O1)	Mode-covering MLE	objective	theorem	MLE
(O2)	Exposure bias	objective	weak empirical	objective (
(O3)	Likelihood $\neq$ correctness	objective	true and central	objective
(T1)	$(1-\varepsilon)^T$ compounding	trained behavior	valid under premises; refuted with a verifier	blind rollou
(T2)	Reversal curse	trained behavior	robust empirical	directional
(T3)	Hallucination is inevitable	trained behavior	weak (strong form refuted by abstention + retrieval)	MLE + op
(T4)	Lipschitz–margin brittleness	trained behavior	robust empirical + theory link	unconstrai
(E1)	Data wall	external	robust empirical (resource curve)	data-bounc

(A1) Softmax bottleneck. Yang et al. (2018) show that a softmax over a d-dimensional hidden state can only produce a stochastic matrix of context-conditional probabilities whose log-probability matrix has rank at most $d+1$. If natural language has higher rank in this sense — as their empirical analysis suggests — the model class is misspecified by construction. Mixture-of-softmaxes restores rank; Kanai et al. (2018) and Chang and McCallum (2022) confirm the predicted improvement and identify mechanisms behind it. This is a theorem. It indicts the softmax parameterization, not the autoregressive factorization.

(A2) Rank collapse. Dong et al. (2021) prove that without residuals, MLPs, or layer normalization, a stack of self-attention layers contracts representations to a rank-one matrix at a doubly-exponential rate in depth. The proof identifies the row-stochastic attention matrix as a Perron–Frobenius contraction on the simplex. The theorem is exact for pure attention. In trained transformers the effect is observed but masked by skip connections and feedforward blocks (Noci et al., 2022; He et al., 2023); whether residual streams alone are enough remains an open empirical question. For the present paper, the cleanest statement is: pure attention provably collapses; the contribution of residual stream depth to recovery is quantitative and architecture-specific.

(A3) Local partition function tax. Softmax attention is, by Ramsauer et al. (2021), a single gradient step on a continuous Hopfield energy whose exponential form was chosen because it admits a closed-form normalizer. Most downstream uses of the attention layer need only *differences* between scores, in which case the local Z cancels in expectation. Whether the per-layer normalization is necessary — as opposed to being a convenient bookkeeping device — is itself a falsifiable architectural claim, and is exactly what sigmoid attention (Wortsman et al., 2023; Ramapuram et al., 2024) and energy heads (Grathwohl et al., 2020) probe. We mark this claim as "established" rather than "theorem" because what is established is the cost, not the necessity.

(A4) Fixed compute ceiling. A standard transformer with depth L, width w, and head count h performs a bounded number of sequential operations per generated token. Merrill and Sabharwal (2023); Strobl et al. (2024) formalize this within circuit complexity: transformers without intermediate steps are placed in fixed-depth uniform classes that cannot solve problems requiring more sequential depth. Chain-of-thought (Wei et al., 2022) and Feng et al. (2023)'s analysis of intermediate steps explicitly use the output token stream to buy more sequential computation. The bound is real. The empirical question is how much chain-of-thought, search, and tool use already buy back, and at what cost in latency.

(O1) Mode-covering MLE. For data distribution p_{data} and model p_θ, the forward KL is

$$D_{\text{KL}}(p_{\text{data}} \,\|\, p_\theta) \;=\; \mathbb{E}_{p_{\text{data}}}[\log p_{\text{data}}(x) - \log p_\theta(x)]\,, \tag{3}$$

which diverges whenever $p_\theta(x) \to 0$ while $p_{\text{data}}(x) > 0$. The optimizer is therefore compelled to spread mass across every observed mode. Theis et al. (2016) and Minka (2005) state this as the standard consequence of forward versus reverse KL; under the noise present in any web-scale corpus, the model assigns nonzero mass to contradictions and incorrect continuations by construction. This is a theorem about the objective. It indicts MLE and, indirectly, the family of methods that use MLE as a pretraining proxy without correction — though instruction tuning and reinforcement learning from verifier feedback are exactly the correction stage.

4.2 The $(1-\varepsilon)^T$ argument, carefully

The $(1-\varepsilon)^T$ argument is the most-cited critique of autoregressive language models in popular discussion. We dwell on it because the gap between its mathematical validity and its actual reach is the main case study for layer discipline in this paper.

The argument is exact under three explicit premises. Let X_t denote the model's t-th generated token and $X_t^\star$ denote the (multi-set of) correct alternatives.

- *Premise 1 (constant rate).* There exists $\varepsilon \in (0,1)$ such that $\mathbb{P}(X_t \neq X_t^\star) = \varepsilon$ for every t.
- *Premise 2 (independence).* The events $\{X_t = X_t^\star\}_{t=1}^T$ are mutually independent.
- *Premise 3 (unrecoverability).* Once $X_t \neq X_t^\star$, the trajectory cannot recover; equivalently, "correct" means literal token-wise match against a single reference path.

Under premises 1–3, $\mathbb{P}[\text{trajectory correct}] = \prod_t (1-\varepsilon) = (1-\varepsilon)^T$. For any $\varepsilon > 0$, this decays exponentially in T (LeCun, 2022; Bachmann and Nagarajan, 2024).

The math is correct. The three premises do the work.

Premise 1 fails in trained models: the realized per-token error rate depends on context, position, topic, and prompt; some stretches are nearly deterministic and others are unstable (Kalai and Vempala, 2024). A constant-rate model is an envelope calculation, not a description of the empirical process.

Premise 2 also fails. Errors cluster. A wrong token typically conditions a region of corrupted state in which subsequent tokens are more, not equally, likely to be wrong. For an actual i.i.d. claim one would have to control for context, and the literature offering such controls reports a context-dependent error rate rather than independent errors (Stechly et al., 2024; Lin et al., 2024).

Premise 3 is the most consequential. A wrong token is unrecoverable only if the surrounding system has no mechanism to detect, reject, or revise it. Modern systems are designed precisely around the opposite assumption. A compiler rejects a syntactically invalid program; a Lean kernel rejects an invalid proof step (de Moura and Ullrich, 2021); a unit-test harness flags a behavioral failure; a search procedure can branch and backtrack; a verifier-in-the-loop trainer reinforces only the trajectories that survive the checker (Lightman et al., 2024; Uesato et al., 2022). A formal-methods workflow can sit in a generate–check–revise loop that turns a long trajectory into a sequence of short, verifier-bounded segments.

The interesting empirical object is not $\mathbb{P}$[token locally wrong]. It is closer to

$$\mathbb{P}[\text{a consequential error survives detection, recovery, and verification}]. \tag{4}$$

That quantity can be much smaller than ε, especially in domains with a hard checker. It can also still be too large. The point is that Equation (4) is a different object from $(1-\varepsilon)^T$. The compounding bound applies to blind rollout. It does not apply to systems whose construction is the negation of premises 1–3.

The companion suite (`experiments/02_ar_pros_cons/probes/probe_06_error_compounding.py`) pre-registers a recoverable-Markov alternative model whose null is geometric decay and whose alternative is a slower-than-geometric process with verifier-bounded segments. Whether the realized rate in formal-methods generation is geometric is the empirical question we explicitly defer to that suite.

4.3 Robust empirical and weak empirical findings

(T2) Reversal curse (robust empirical). Berglund et al. (2024) demonstrate that models trained on "A is B" do not reliably retrieve "B is A". The result reproduces across model scales and across structured-relation tests. The indicted target is the directional next-token objective combined with static text training, not the autoregressive factorization itself — as the experimental controls show, conditioning the same model on a paraphrase that recovers symmetric structure can lift performance.

(T4) Lipschitz–margin (robust empirical + theory link). Tsuzuku et al. (2018) and Virmaux and Scaman (2018) give the explicit bound and the upper estimate $L \leq \prod_i \|W_i\|_2$. Adversarial robustness studies (Szegedy et al., 2014; Goodfellow et al., 2015; Carlini et al., 2023) report small input perturbations that swing model outputs across decision boundaries despite high confidence. The mechanism is consistent: confidence and geometric margin are decoupled. The indictment is unconstrained Lipschitz training rather than autoregression.

(T3) Hallucination (weak in its strong form). Kalai and Vempala (2024) prove that calibrated next-token training places nonzero mass on incorrect continuations and conclude that an open-generation system trained this way will hallucinate at a nonzero rate. The strong version of the claim — hallucination is *inevitable* — requires also that the system not abstain, not retrieve, and not check. Kandpal et al. (2023) and Maynez et al. (2020) document realized hallucination patterns; Liu et al. (2024) document grounding effects. We take the conservative reading: the structural pressure follows from the objective, but the realized rate is a system property that retrieval, abstention, and verification can change (Ji et al., 2023).

(O2) Exposure bias (weak empirical). Scheduled sampling (Bengio et al., 2015) and reinforcement-learning extensions (Ranzato et al., 2016) demonstrated the mechanism on small models. On large models, the realized gap between teacher-forced training and free-running generation appears smaller (Wang and Sennrich, 2020), though the question is not closed for very long horizons. We flag this as weak rather than refuted because its magnitude in the long-horizon regime is precisely the regime where it would matter most, and the existing evidence is concentrated in short-horizon benchmarks.

(E1) Data wall (robust empirical, resource curve). The argument depends only on (i) the empirical scaling law that loss requires exponential data for linear improvement and (ii) an estimate of available high-quality text (Villalobos et al., 2024). Neither piece requires a "model cannot do X" claim; both are observed. The companion suite includes a probe that re-fits the scaling law on VeriBench subsets (Daruru et al., 2026) and extrapolates.

4.4 Explicitly excluded as not load-bearing

Two further objections appear regularly and are not load-bearing in our reading.

The "stochastic parrot" framing (Bender et al., 2021) is not operationalizable in its general form: it makes no falsifiable prediction at the granularity at which it is stated. Its operationalizable pieces (reversal, planning, compositional generalization) reduce to claims we have already addressed.

The claim that autoregressive factorization *itself* caps expressivity is mostly false. The chain rule of probability decomposes any joint over a finite sequence into left-to-right conditionals. Practical bottlenecks come from the parameterization of the conditional, the objective used to train it, and the compute used to run inference — not from the existence of the factorization.

4.5 What this leaves

The surviving load-bearing claims separate cleanly into three families:

1. *Against the softmax / local normalization:* (A1), (A2), (A3).
2. *Against the MLE objective:* (O1).
3. *Against fixed-compute generation:* (A4), (T4).

The next three sections take this as input and ask, for each family, what alternative architectures put in its place. The recurring question is whether the alternative pays the same cost in a different place or absorbs it.

5 Energy-Based Models and the Partition Function

The promise of energy-based models is to address the most central limitation that survived Section 4: local next-token likelihood is not global correctness ((O3)). An energy-based model defines a scalar energy on whole objects and turns global compatibility into a single scoring function. The cost is that exact normalization, exact likelihood, and exact sampling become hard. This section states the formulation, the central obstacle, and the reframing of softmax attention as an energy-descent step.

5.1 Energy and probability

An energy-based model parameterizes a scalar function $E_\theta : \mathcal{X} \to \mathbb{R}$ on the object space $\mathcal{X}$ and induces the Boltzmann distribution

$$p_\theta(x) \;=\; \frac{\exp(-E_\theta(x))}{Z_\theta}, \qquad Z_\theta \;=\; \int_{\mathcal{X}} \exp(-E_\theta(x))\,\mathrm{d}x. \tag{5}$$

For finite $\mathcal{X}$, the integral becomes a sum. The energy can be any neural network producing a scalar; Z_θ is the partition function (LeCun et al., 2006; Song and Kingma, 2021).

The appeal is that E_θ can be parameterized in any way that respects the geometry of the problem, free of the constraint that its output be a normalized probability. A model that scores a whole sequence, a whole image, or a whole proof can be defined directly:

$$E_\theta(x_{1:T}) \;=\; f_\theta(x_{1:T}), \tag{6}$$

where f_θ is, for example, a transformer that returns a single scalar from a sequence input (Grathwohl et al., 2020; Du and Mordatch, 2019). Low energy denotes high compatibility; high energy denotes incompatibility. Two equally good completions of a prompt can have equal energy without competing for probability mass in the way they would in a locally normalized model.

5.2 The partition function on sequences

Equation (5) is honest about its cost. For a discrete sequence space $\mathcal{X} = \mathcal{V}^T$,

$$Z_\theta \;=\; \sum_{x \in \mathcal{V}^T} \exp(-E_\theta(x)), \tag{7}$$

which contains $|\mathcal{V}|^T$ terms. For a 32,000-token vocabulary and a 512-token sequence, that is a sum of 32000^{512} terms, well beyond enumeration. For continuous sequence embeddings $\mathcal{X} = (\mathbb{R}^d)^T$,

$$Z_\theta \;=\; \int_{(\mathbb{R}^d)^T} \exp(-E_\theta(x))\,\mathrm{d}x, \tag{8}$$

which is a high-dimensional integral whose only general guarantee is that it is hard (Murphy, 2023).

The autoregressive factorization in Equation (2) avoided this by paying T local normalizers of size $|\mathcal{V}|$ each instead of one global normalizer of size $|\mathcal{V}|^T$. That is the structural trade. The cost of the autoregressive bargain is the per-token mode-covering objective ((O1)) and the local-then-global mismatch ((O3)). The cost of the energy-based bargain is Equations (7) and (8) or whatever approximation stands in for them.

The natural taxonomy of training methods for energy-based models reflects this: each method makes a different choice about whether to approximate Z_θ directly, replace it with a tractable surrogate, learn only its gradient, or avoid it altogether at training time. Section 6 surveys these. The natural taxonomy of inference methods reflects the same question for sampling and posterior queries; Section 7 surveys those.

5.3 The Hopfield reframing

A reframing that pre-dates much of the EBM discussion makes the gap between autoregressive and energy-based language modeling narrower than slogans suggest. Ramsauer et al. (2021) prove that scaled-dot-product attention, applied to query $\xi \in \mathbb{R}^d$ and stored patterns $X \in \mathbb{R}^{N\times d}$, is one gradient step on the continuous Hopfield energy

$$E(\xi) \;=\; -\mathrm{lse}(\beta\, X\xi) + \tfrac{1}{2}\xi^\top\xi + c, \tag{9}$$

with update

$$\xi_{\text{new}} \;=\; X\,\mathrm{softmax}(\beta\, X\xi). \tag{10}$$

The match is literal for a single attention head with $K = V$, with the standard $1/\sqrt{d_k}$ scaling absorbed into β; Ramsauer et al. (2021) extend it to the $K \neq V$ case by a substitution. Capacity in this energy is exponential in pattern dimension (Demircigil et al., 2017).

The reframe is consequential.

- Softmax attention is *already* an energy-descent step. The research question is which energy and how many steps.
- The local partition function (A3) is the part of lse that makes the energy admit a tractable closed-form step. Replacing lse with something exp-free typically removes the global competition, injective query-to-weight map, or spikiness properties that make softmax attention work (Zhang et al., 2024b; Han et al., 2024).
- The full transformer is not the gradient of any single energy. Multi-layer composition with layer norm, MLPs, and residuals takes the equivalence beyond literal at the layer-stack level. It is sound to call a single attention head an energy-descent step; it is metaphor to call the full transformer "minimizing a single energy."

This bounds the claims a project pitched against softmax can credibly make. An EBM design that wins by "replacing softmax with an energy" has to specify which property of Equation (9) the replacement preserves and which it abandons. Recent post-mortem and synthesis papers (Jelassi et al., 2024; Zhang et al., 2024b; Han et al., 2024; Mongaras and Larson, 2025) consistently arrive at "softmax plus modification" rather than "no softmax," which is consistent with the energy reading.

The remaining question — the one that the rest of the paper takes seriously — is not "EBM versus AR" framed as a binary. It is which energy, where it is evaluated, and whether Z_θ can be amortized or avoided at training and at inference.

6 Training Energy-Based Models

The training methods for energy-based models can be organized by what they do with Z_θ. Table 2 previews the taxonomy. We then take each method in turn.

Table 2: Training methods for energy-based models, organized by their treatment of the partition function. "Cost moved to" indicates where the difficulty appears in practice.

Method	Treatment of Z_θ	Cost moved to
MLE with sampling	estimated via MCMC samples	MCMC mixing per gradient step
Contrastive divergence	truncated MCMC chains	bias from short chains
Persistent CD	long-lived chains across updates	chain decorrelation
Noise-contrastive estimation	cast as binary classification	quality of the noise distribution
Score matching	avoids Z_θ entirely	second-order derivatives of the energy
Denoising score matching	avoids Z_θ entirely	noise schedule design
Sliced score matching	avoids Z_θ entirely	random-projection variance
Ratio matching	avoids Z_θ entirely (discrete)	per-coordinate ratio comparisons
Stein discrepancy	avoids Z_θ entirely	kernel choice
Implicit differentiation / JEM	shares Z_θ with a classifier	inference for the gradient

6.1 Maximum likelihood with sampled gradients

The gradient of the negative log-likelihood under Equation (5) is

$$-\nabla_\theta \log p_\theta(x) \;=\; \nabla_\theta E_\theta(x) \;-\; \mathbb{E}_{x' \sim p_\theta}\big[\nabla_\theta E_\theta(x')\big]\,. \tag{11}$$

The data term $\nabla_\theta E_\theta(x)$ is direct. The model expectation is the part that requires samples from p_θ, typically via Markov chain Monte Carlo (Hinton, 2002; Carreira-Perpiñán and Hinton, 2005). Each gradient step incurs the cost of running a sampler until it has approximately mixed. For energy parameterizations as expressive as a transformer, sampling does not mix quickly (Nijkamp et al., 2020; Du and Mordatch, 2019), and exact MLE becomes operationally a budgeted approximation rather than a literal objective.

6.2 Contrastive divergence and its persistent variant

Hinton (2002) introduced contrastive divergence (CD-k), which replaces the long MCMC run in Equation (11) with a short chain of k steps starting from a data point. The resulting estimator is biased but cheap. Persistent contrastive divergence (Tieleman, 2008) maintains chains across parameter updates so that the negative samples track a slowly-changing p_θ. Du and Mordatch (2019) and Nijkamp et al. (2020) apply these to deep image energies with sampler buffers. The empirical pattern is that CD trains and stabilizes only when the energy landscape is regular enough for short chains to be informative; for very expressive energies, the bias of CD becomes the limiting factor and persistent variants are typically used.

6.3 Noise-contrastive estimation

Noise-contrastive estimation (Gutmann and Hyvärinen, 2010, 2012) converts density estimation into binary classification: distinguish data samples from samples drawn from a known noise distribution p_n. The unnormalized energy enters the classification logit, and $\log Z_\theta$ becomes a learnable scalar parameter trained jointly. Asymptotically, the classifier learns $\log(p_{\text{data}}/p_n)$, and the recovered density is correct up to the noise distribution's coverage of the data (Ma and Collins, 2018). The cost moves from sampling p_θ to choosing p_n. A noise distribution that misses regions of the data renders the classifier blind to them (Gutmann et al., 2014). Conditional NCE (Mnih and Teh, 2012) adapts the method to language modeling by sampling negatives from a unigram base.

6.4 Score matching

Score matching (Hyvärinen, 2005) avoids Z_θ entirely by matching $\nabla_x \log p_\theta(x)$ to $\nabla_x \log p_{\text{data}}(x)$ in expectation. Because the gradient of $\log Z_\theta$ with respect to x is zero, the score depends only on E_θ:

$$\nabla_x \log p_\theta(x) \;=\; -\nabla_x E_\theta(x). \tag{12}$$

The original Fisher-divergence loss requires the trace of the Hessian of E_θ, which is expensive for deep networks. Two reformulations sidestep this.

Denoising score matching. Vincent (2011) prove that score matching against the smoothed density $\widetilde{p}(\tilde{x}) = \int q_\sigma(\tilde{x} \mid x) p_{\text{data}}(x) \mathrm{d}x$ for a known noise kernel q_σ reduces to predicting the noise $\nabla_{\tilde{x}} \log q_\sigma(\tilde{x} \mid x)$. This is the loss now standard in diffusion models (Song and Ermon, 2019; Ho et al., 2020).

Sliced score matching. Song et al. (2020) replace the Hessian trace with random projections, giving an unbiased estimator with a single backward pass per sample.

Score matching has been adapted to text via discrete-variable extensions and to mixed sequence-embedding settings (Meng et al., 2022; Lou et al., 2024). The price of avoiding Z_θ at training is that the model is implicit in the score, and sampling requires running an iterative procedure (Section 7.1).

6.5 Ratio matching and discrete energies

For discrete variables, the derivative-based score-matching identities do not apply. Hyvärinen (2007) introduce ratio matching: for each datum and each coordinate, the model matches the ratio of p_θ between flipping and not flipping the coordinate. Generalized score matching for discrete data (Lyu, 2009; Grathwohl et al., 2021) extends this to wider notions of local perturbation. In the language modeling context, Deng et al. (2020) and Lou et al. (2024) use related ideas to define energies whose training loss is a coordinate-wise comparison rather than a full normalization.

6.6 Joint energy / classifier hybrids

Grathwohl et al. (2020) observe that a logit-producing classifier defines an unconditional density via $\log p(x) = \mathrm{lse}(\mathrm{logits}(x))$ up to a constant. Treating the classifier as both a discriminative model and an EBM gives the joint energy-based model (JEM). Training combines the standard cross-entropy classification loss with a generative term computed via MCMC, sharing the partition function with the classification head. The mechanism is consistent with the Hopfield reframe in Section 5.3: a softmax-headed classifier is already an EBM.

6.7 Stein and kernelized objectives

Kernelized Stein discrepancy (Liu et al., 2016; Chwialkowski et al., 2016) provides an unnormalized goodness-of-fit objective that uses only $\nabla_x \log p_\theta(x)$. Training an energy by minimizing the empirical Stein discrepancy avoids Z_θ at the cost of choosing a kernel and accepting bias proportional to the kernel's ability to detect mass discrepancies.

6.8 What the training survey leaves

Each method in Table 2 pays the cost of Z_θ somewhere. Score-matching, denoising, and Stein methods push it into inference, where sampling becomes an iterative procedure (next section). NCE pushes it into the choice of noise distribution and into the asymptotic bias if the noise misses modes. MCMC-based estimators pay it directly in mixing time, which is the binding constraint on training large discrete energies.

The honest comparison with autoregressive language modeling is therefore not "EBM versus AR." It is "where does the cost of Z_θ go, and is the destination cheaper than T local softmaxes plus the mode-covering objective?" For domains with a hard verifier — including formal-methods generation, code, structured outputs, and constraint satisfaction — the cost of an inference-time iterative procedure can be amortized across a small number of verifier-bounded segments, which is precisely the setting in which the partition-function obstacle is most plausibly evaded in practice. We return to this in Section 9.

7 Inference in Energy-Based Models

Sampling from Equation (5) or computing posterior queries against it is the second place the partition function bites. This section organizes inference methods by the discreteness of the object and by whether the model exposes its score, its energy, or only its conditional updates.

7.1 Continuous: Langevin dynamics and Hamiltonian methods

Langevin dynamics defines an SDE whose stationary distribution is p_θ:

$$\mathrm{d}x \;=\; -\nabla_x E_\theta(x)\,\mathrm{d}t \;+\; \sqrt{2}\,\mathrm{d}W_t, \tag{13}$$

discretized as $x_{k+1} = x_k - \tau\nabla_x E_\theta(x_k) + \sqrt{2\tau}\,\eta_k$ for noise $\eta_k \sim \mathcal{N}(0, I)$ and step $\tau > 0$. Convergence to p_θ holds under mild regularity (Roberts and Tweedie, 1996). The cost of inference is the number of iterations to traverse a representative region of the energy landscape. Stochastic gradient Langevin dynamics (Welling and Teh, 2011) couples Equation (13) with stochastic gradient updates to scale. Hamiltonian Monte Carlo (Neal, 2011; Betancourt, 2017) augments with momentum, reducing the number of effective steps when the energy is smooth.

Score-based generative models (Song and Ermon, 2019; Song et al., 2021b) use a noise-conditional score network in place of $\nabla_x E_\theta$ and integrate Equation (13) along a denoising schedule. The training methods of Section 6.4 return an estimate of the score that this inference loop consumes.

7.2 Discrete: Gibbs, simulated annealing, and local-search analogues

For discrete sequence spaces, Langevin dynamics does not directly apply. Gibbs sampling (Geman and Geman, 1984) updates one coordinate at a time conditional on the rest; the conditional itself requires evaluating $|\mathcal{V}|$ energies per step. Block-Gibbs and Metropolis–within-Gibbs variants trade per-step cost for mixing rate. For energies parameterized as deep networks over discrete sequences, recent work (Grathwohl et al., 2021; Emami et al., 2024) proposes continuous relaxations and gradient-guided proposals that accelerate mixing without losing the discrete target.

Simulated annealing (Kirkpatrick et al., 1983) traces the same energy under a temperature schedule that contracts as iteration proceeds. It is a natural model for "inference as optimization" when only the MAP is required.

7.3 Variational inference and the free-energy view

Variational methods replace exact inference with a tractable surrogate $q_\phi(z)$ over latents z and minimize the variational free energy

$$\mathcal{F}[q_\phi] \;=\; \mathbb{E}_{q_\phi}[E_\theta(z)] \;-\; H(q_\phi), \tag{14}$$

where $H(q_\phi)$ is the entropy. Minimization of Equation (14) gives the closest q_ϕ to p_θ in reverse-KL (Wainwright and Jordan, 2008). Rezende et al. (2014); Kingma and Welling (2014) introduce amortized variants that learn q_ϕ as a neural network and have become the standard scaffold for variational autoencoders. Ranganath et al. (2014) make the approach black-box-applicable. The mean-field assumption that q_ϕ factorizes across coordinates is the analogue of independent token modeling and is what attention's row-stochastic structure already implements (Ramsauer et al., 2021).

7.4 Posterior queries: conditional sampling and amortization

Many practical EBM uses do not need samples from the unconditional p_θ. They need samples from $p_\theta(x \mid c)$ for a conditioning context c — a prompt, a partial program, a goal state. Du et al. (2023) demonstrate that conditional energy descent is competitive with autoregressive sampling on structured outputs, and that energy-based composition can be applied in latent space. For tasks with a hard verifier, conditional sampling on the verifier's accept set reduces to constrained energy descent: project at each step onto the verifier-bounded subspace (Cobbe et al., 2021).

7.5 Energy descent at inference and adaptive compute

The reason inference deserves a dedicated section is the claim that motivated the field in the first place: *the number of inference steps is a free parameter*, and energy-based inference can spend more on harder inputs. This is the architectural counterpart to chain-of-thought. Where chain-of-thought converts more inference into more tokens, energy descent converts more inference into more iterations of the same forward pass (LeCun, 2022; Du et al., 2023).

The empirical question is whether $K > 1$ steps of energy descent on a fixed model are ever more compute-efficient than the equivalent additional training compute spent on a one-step model. For well-separated patterns under Equation (9), Ramsauer et al. (2021) prove that one step is already optimal. For not-well-separated inputs — precisely the inputs we care about — multiple steps are expected to help, and partial results in this direction exist (Bachlechner et al., 2021; Bansal et al., 2022; Schwarzschild et al., 2021). The companion suite (`experiments/02_ar_pros_cons/probes/probe_05.py`) explicitly tests whether iteration is compute-cheaper than parameter count in a verifier-bounded setting.

7.6 What the inference survey leaves

Inference in energy-based models is a budgeted decision: how many iterations, with what proposal, against which conditioning set, with what verifier hook. The autoregressive recipe makes the equivalent decision implicitly — one forward pass, beam search, or sample-then-verify — but does so with a fixed compute budget per output token. The interesting comparison is not "EBM inference is hard" versus "AR inference is easy." It is "where do you want to spend the variable-compute budget," phrased clearly enough that the trade is visible. Section 8 broadens the comparison to alternatives that occupy intermediate positions on this trade.

8 Alternatives Beyond Energy-Based Models

Energy-based models are not the only alternative. The space of design choices for sequence generation is larger, and several lines of work occupy intermediate positions between the autoregressive recipe and a fully energy-based one. We organize them by which design choice they replace and what they pay in exchange.

8.1 Diffusion and denoising models

Diffusion models (Sohl-Dickstein et al., 2015; Ho et al., 2020; Song et al., 2021b; Nichol and Dhariwal, 2021) define a forward Markov chain that progressively destroys structure and a learned reverse chain that progressively restores it. Training reduces to denoising score matching (Section 6.4) along a fixed noise schedule. The resulting model is an EBM in disguise: the learned reverse process is the score of a sequence of perturbed densities (Song et al., 2021a).

Continuous data. For images, audio, and continuous embeddings, the inference cost is the number of denoising steps; distillation (Salimans and Ho, 2022; Song et al., 2023) compresses this from hundreds of steps to a small number.

Discrete data. For text and structured outputs, discrete and masked diffusion (Austin et al., 2021; Hoogeboom et al., 2021; Lou et al., 2024; Nie et al., 2024) replace Gaussian perturbation with token-level corruption or masking. The training loss remains a denoising objective; inference is iterative refinement on a partially-masked sequence. The relevant trade against the autoregressive recipe is that diffusion trains and samples without the directional constraint of left-to-right, which has implications for the reversal curse ((T2)) and for the ability to amortize global constraints across iterations.

8.2 Flow matching and continuous normalizing flows

Flow matching (Lipman et al., 2023; Liu et al., 2023; Albergo and Vanden-Eijnden, 2023) replaces the Gaussian diffusion path with a learned vector field whose integration produces samples. Continuous normalizing flows (Chen et al., 2018; Grathwohl et al., 2019) preserve exact likelihood by integrating the change-of-variables formula along the field. Both inherit the inference-time iteration of diffusion; both avoid the partition function by parameterizing transport rather than density.

8.3 Joint-embedding predictive architectures

LeCun (2022) proposes joint-embedding predictive architectures (JEPA) that replace pixel- or token-level reconstruction with feature-space prediction. The energy is the distance between predicted and target embeddings. Assran et al. (2023) and Bardes et al. (2024) demonstrate the recipe on image and video. The position is that reconstruction-level prediction allocates capacity to perceptually irrelevant details and that predicting in an abstracted feature space is a more efficient use of the same data. The training methods inherit the contrastive–vs–non-contrastive distinction familiar from self-supervised learning (Chen et al., 2020; Grill et al., 2020; He et al., 2022); the partition function does not directly appear because the loss is a feature-space distance rather than a normalized density.

8.4 State-space models and recurrent alternatives

State-space models (Gu et al., 2022; Gu and Dao, 2024; Dao and Gu, 2024) replace softmax attention with an input-dependent linear recurrence whose state is fixed-dimensional. The forward pass is a streaming linear-time operation; training admits a parallel-prefix-scan formulation. The targeted indictment is the per-layer partition function tax ((A3)). The cost is the fixed-state recall limit ((A5)): Jelassi et al. (2024) show that any fixed-dimensional recurrent model cannot copy strings of arbitrary length, where a two-layer softmax transformer can. Hybrids (Lieber et al., 2024; Dao and Gu, 2024; DeepSeek-AI, 2026) interleave linear recurrence with softmax attention precisely to recover the recall property while keeping most of the per-layer cost linear. The post-mortem reading (Zhang et al., 2024b; Han et al., 2024; Mongaras and Larson, 2025) is that the load-bearing property of softmax is global competition over a non-injective domain, and any replacement must preserve it at some scale.

8.5 Latent-variable and variational architectures

Variational autoencoders (Kingma and Welling, 2014; Rezende et al., 2014) factor the joint as $p(x, z) = p(z)p(x \mid z)$ and train both factors against an evidence lower bound that uses a tractable surrogate posterior $q_\phi(z \mid x)$. The partition function over x does not appear; the latent prior $p(z)$ is typically chosen tractable. For sequence modeling, latent-variable architectures have been competitive with autoregressive models in restricted regimes (Razavi et al., 2019; Vahdat et al., 2021) but have not displaced them at the scale of public language models, in part because the dense per-token supervision of the autoregressive recipe is hard to match with a single latent code per object.

8.6 Autoregressive models with verifiers, search, and tools

The most empirically successful response to the surviving objections of Section 4 has been to keep the autoregressive recipe and surround it with mechanisms that violate the premises of the strong critiques.

Verifier-in-the-loop training. Process- and outcome-based reward models (Lightman et al., 2024; Uesato et al., 2022; Wang et al., 2024) learn from a verifier's accept/reject signal. Reinforcement learning from verifier feedback (Cobbe et al., 2021; Gao et al., 2023a) aligns the trained model against the same signal. The targeted indictment is mode-covering MLE ((O1)) and the unrecoverability premise of error compounding ((T1)).

Search and self-consistency. Self-consistency decoding (Wang et al., 2023), Monte Carlo tree search over reasoning trees (Yao et al., 2023; Hao et al., 2023; Zhang et al., 2024a), and verifier-guided sampling (Chen et al., 2024) buy variable compute at inference by running multiple trajectories and selecting the one a learned or external scorer prefers.

Retrieval-augmented generation. Retrieval-augmented generation (Lewis et al., 2020; Izacard et al., 2023; Khattab et al., 2022) couples the generator to a retriever over an external corpus, addressing the structural pressure toward plausible completion that drives hallucination ((T3)).

Tool use and program synthesis. Schick et al. (2023); Chen et al. (2023); Gao et al. (2023b) delegate computation to external interpreters, calculators, search engines, and verifiers. The autoregressive model writes the program; the program is executed elsewhere. Each of these mechanisms is a small step toward the broader principle that load-bearing computation belongs outside the language model.

The honest summary is that the most successful production system today is an autoregressive language model wrapped in mechanisms that explicitly negate (T1)'s third premise (unrecoverability), implicitly damp (T3)'s rate via retrieval and verification, and externalize part of (A4) via search. A clean comparison with an energy-based architecture has to specify which of these mechanisms it retains, which it absorbs, and which it discards.

8.7 Hopfield, modern associative memory, and the architectural middle ground

The Hopfield reframe (Section 5.3) is itself an architectural middle ground. Ramsauer et al. (2021) and follow-up work (Millidge et al., 2022; Burns et al., 2024) treat the transformer as a sequence of associative memory retrievals. Project proposals that propose "different energies" under this view (Krotov and Hopfield, 2021; Hoover et al., 2023) are not opposing the transformer; they are proposing different per-step energies for the same recurrence pattern. Whether any of these designs simultaneously preserves the five properties identified as load-bearing in Zhang et al. (2024b); Han et al. (2024); Mongaras and Larson (2025) — global competition, injective query-to-weight map, low-entropy attention, streaming-stable kernel, and exponential storage capacity — remains an open empirical question. The companion repository's `docs/softmax_graveyard.md` pre-registers a five-property accounting that any proposal must address.

9 Where the Hard Part Should Live

This paper organized the standard objections to autoregressive language models, separated them into three logically independent indictments, evaluated each against a clear bar, and surveyed the alternatives that target each. A summary of the position the paper supports is below.

What survives, and what it indicts. Three families of objections survive the audit of Section 4. The softmax bottleneck, doubly-exponential rank collapse in pure attention, and the local partition function tax indict the parameterization of the per-step distribution and the per-layer attention. The mode-covering pressure of forward-KL training indicts the maximum-likelihood objective on noisy corpora. The fixed-compute ceiling and the Lipschitz–margin gap indict the inference procedure of single-pass blind rollout. The $(1-\varepsilon)^T$ argument is mathematically valid and rhetorically useful, but its three premises are explicitly violated by verifier-augmented systems and by feedback loops with hard checkers; treating it as an impossibility proof confuses a warning label for a theorem.

Where the alternatives put the hard part. Energy-based models address the local-then-global mismatch by scoring whole objects. The cost is the partition function or its approximation, paid either at training time (MLE with MCMC, contrastive divergence), at inference time (score-matching plus Langevin or diffusion sampling), or at neither but through a noise distribution whose coverage becomes the bottleneck (NCE). Diffusion and flow-matching models share the inference-time iteration of score-matching; they have largely succeeded in continuous data and are an active question in discrete data. Joint-embedding predictive architectures avoid token-level reconstruction and the partition function but inherit the contrastive-versus-non-contrastive design space of self-supervised representation learning. State-space models target only the per-layer partition function and, in their pure form, inherit the fixed-state recall limit; hybrid stacks restore softmax attention at some scale and recover the property. Autoregressive models with verifiers, search, retrieval, and tools attack the third indictment directly by negating the unrecoverability premise and the fixed-compute premise, and are the most empirically successful production response.

What this paper deliberately defers. This paper is the literature-grounded half of a two-part project. The empirical claims — which mechanisms are load-bearing inside a trained model on a domain with a hard verifier, and how the realized error process compares to the geometric prediction of Section 4.2 — are deferred to the companion suite, `experiments/02_ar_pros_cons`, which pre-registers a probe-by-probe accounting at the correct layer for each surviving claim. We will not commit to a verdict on the bridge between isolated mechanism strength and downstream realized effect until that suite reports.

What we are willing to say now. The honest summary of the literature is that the most-discussed critique of autoregressive language models, $(1-\varepsilon)^T$, is the weakest one in the surviving set, and the most-discussed alternative, "replace softmax with an energy," is, by the Hopfield reading, a question of *which* energy rather than *whether* an energy. The load-bearing research questions are narrower and more concrete than the slogans suggest:

1. For each load-bearing isolated mechanism in Table 1, what is its realized effect inside an end-to-end trained model on a hard-verifier domain?
2. For each candidate energy, which of the five load-bearing softmax properties does it preserve, and at what cost?

3. Under what conditions is $K > 1$ steps of energy descent at inference cheaper than the equivalent additional training compute?
4. What is the empirical decay law of $(1 - \varepsilon_{\text{effective}})^T$ for verifier-augmented generation on Lean, and is it geometric, sub-geometric, or polynomial?

Each of these is a measurement question with a stated null and a stated alternative. The companion suite is structured to answer them. The answer will determine whether the architectural questions this paper raises remain open or are resolved by ablation.

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A Open Questions

This appendix records the precise open questions that the paper raises but does not resolve. Each is stated as a measurement with a null and an alternative.

A.1 Mechanism-level open questions

Q1: Is rank collapse (A2) masked in trained transformers? Null: a depth-L pre-norm transformer trained on standard data exhibits the same row-rank statistics as a randomly initialized one of the same depth and width. Alternative: residual streams plus MLPs interrupt the contraction enough that the row-rank gap with random initialization grows with L.

Q2: Is the local partition function (A3) removable on text? Null: swapping softmax attention for sigmoid attention with norm correction matches downstream task loss within confidence intervals at matched compute. Alternative: the per-layer normalizer is load-bearing for content-addressed retrieval on long context (Zhang et al., 2024b; Han et al., 2024).

Q3: Does $K > 1$ steps of energy descent at inference beat equivalent additional training compute? Null: a one-step energy model trained with C FLOPs matches the test loss of a K-step energy model trained with C/K FLOPs and run for K steps at inference. Alternative: iteration buys more than additional training on inputs whose patterns are not well-separated.

A.2 $(1-\varepsilon)^T$-shaped questions

Q4: Is the per-token error rate constant in a trained model? Null: in a held-out generation set, the per-position error rate is independent of position and prior errors. Alternative: the error rate is context-dependent and clusters along trajectories.

Q5: Is the surviving-error rate of verifier-augmented Lean generation geometric in T? Null: $\log \mathbb{P}[\text{trajectory accepted by Lean kernel}]$ is linear in trajectory length with a constant slope. Alternative: the slope is shallower than $\log(1-\varepsilon_{\text{single-step}})$ by a verifier-induced gap whose magnitude is the quantity of interest.

Q6: Is the reversal failure (T2) closed by any-order training? Null: a model trained with random ordering of the next-token target inherits reversal failures at the same rate as a left-to-right model. Alternative: any-order training (Sahoo et al., 2024; Kitouni et al., 2024) reduces but does not eliminate the gap.

A.3 Scaling / external open questions

Q7: Is the high-quality public text corpus the binding constraint? Null: the scaling-law fit on VeriBench subsets predicts the realized loss curve at the high end of the public-data range. Alternative: synthetic data and verifier-generated data shift the law before the dataset runs out (Kazemi et al., 2024).

Q8: Do mixture-of-softmaxes heads close the softmax bottleneck (A1) in trained models? Null: mixture-of-softmaxes matches a single softmax at compute parity. Alternative: a non-trivial gap survives, consistent with Yang et al. (2018); Chang and McCallum (2022).

B Derivations Used in the Body

This appendix records the short derivations cited in the body but kept out of the main text for readability.

B.1 Autoregressive normalization is automatic

For local conditionals $p_\theta(x_t \mid x_{<t})$ each normalized over $\mathcal{V}$,

$$\sum_{x \in \mathcal{V}^T} \prod_{t=1}^{T} p_\theta(x_t \mid x_{<t}) = \sum_{x_{<T} \in \mathcal{V}^{T-1}} \prod_{t=1}^{T-1} p_\theta(x_t \mid x_{<t}) \cdot \underbrace{\sum_{x_T \in \mathcal{V}} p_\theta(x_T \mid x_{<T})}_{=1} \tag{15}$$

$$= \cdots = 1. \tag{16}$$

This is the statement that left-to-right factorization with locally normalized conditionals produces a normalized joint by construction, with no global Z_θ.

B.2 Forward-KL is zero-avoiding (mode-covering)

For data distribution p_{data} and model p_θ,

$$D_{\text{KL}}(p_{\text{data}} \,\|\, p_\theta) = \int p_{\text{data}}(x) \log \frac{p_{\text{data}}(x)}{p_\theta(x)} \,\mathrm{d}x \tag{17}$$

$$= -H(p_{\text{data}}) - \int p_{\text{data}}(x) \log p_\theta(x) \,\mathrm{d}x. \tag{18}$$

The integral diverges to $+\infty$ whenever $p_{\text{data}}(x) > 0$ and $p_\theta(x) \to 0$, so any optimizer minimizing the KL is compelled to keep $p_\theta(x) > 0$ on the support of p_{data}. This holds whether p_{data} is the empirical distribution or its smoothed version, and is the source of the mode-covering pressure cited in Section 4.1.

B.3 $(1-\varepsilon)^T$ under the three explicit premises

Let X_t denote the generated token at step t and $X_t^\star$ the (multi-set of) correct alternatives. Under *constant rate* ($\mathbb{P}(X_t \neq X_t^\star) = \varepsilon$), *independence* ($\{X_t = X_t^\star\}_{t=1}^T$ mutually independent), and *unrecoverability* ("correct trajectory" is the literal intersection $\bigcap_t \{X_t = X_t^\star\}$),

$$\mathbb{P}[\text{trajectory correct}] = \mathbb{P}\left[\bigcap_{t=1}^{T} \{X_t = X_t^\star\}\right] \tag{19}$$

$$\overset{\text{indep.}}{=} \prod_{t=1}^{T} \mathbb{P}[X_t = X_t^\star] \overset{\text{const.}}{=} (1-\varepsilon)^T. \tag{20}$$

The premises are the entire content of the proof. Without independence, the product step does not hold; without constant rate, the exponent is replaced by a sum that does not factor; without unrecoverability, the intersection is no longer the event of interest.

B.4 Score-matching gradient does not depend on Z_θ

For $p_\theta(x) = \exp(-E_\theta(x))/Z_\theta$,

$$\nabla_x \log p_\theta(x) = \nabla_x[-E_\theta(x) - \log Z_\theta] \tag{21}$$

$$= -\nabla_x E_\theta(x), \tag{22}$$

since Z_θ is independent of x. The Fisher-divergence loss $\mathbb{E}_{p_{\text{data}}}\left[\frac{1}{2}\|\nabla_x \log p_\theta(x) - \nabla_x \log p_{\text{data}}(x)\|^2\right]$ is therefore expressible as a function of E_θ alone (Hyvärinen, 2005). The original form requires $\nabla_x \log p_{\text{data}}$, which is unknown; Hyvärinen (2005)'s identity rewrites the loss using integration by parts as $\mathbb{E}_{p_{\text{data}}}\left[\frac{1}{2}\|\nabla_x E_\theta(x)\|^2 + \text{tr}(\nabla_x^2 E_\theta(x))\right]$, eliminating dependence on p_{data}'s score. Denoising score matching (Vincent, 2011) replaces the Hessian trace with a tractable denoising target via a known noise kernel.